

TWILIO IVR CALL & SMS CONFIRMATION AUTOMATION

Built with n8n

Assignment Submission Report

Project Type	Voice & SMS Automation
Automation Platform	n8n Cloud
Voice Platform	Twilio Voice API
Messaging Platform	Twilio Programmable Messaging
Communication Format	Twiml
Integration Method	REST APIs & Webhooks

1. Project Overview & Objectives

1.1 Project Overview

The Twilio IVR Call & SMS Confirmation Automation is an end-to-end voice automation workflow built using n8n Cloud and Twilio. The workflow automatically initiates an outbound phone call, plays an Interactive Voice Response (IVR) message, collects the recipient's keypad input using DTMF, processes the response through an n8n webhook, and sends an SMS confirmation when the recipient presses the required key.

The project demonstrates how a workflow automation platform can be integrated with a telecommunications API to create a practical voice-based automation system.

1.2 Project Objectives

- Configure Twilio credentials for API authentication.
- Provision and configure a Twilio phone number.
- Build an outbound voice-call workflow in n8n.
- Use Twilio's Voice API to initiate a call.
- Create an IVR using TwiML.
- Use **<Say>** to provide a text-to-speech greeting.
- Use **<Gather>** to capture DTMF keypad input.
- Receive Twilio webhook requests in n8n.
- Extract the pressed digit, Call SID, and caller information.
- Route the workflow according to the user's selection.
- Send an SMS confirmation through Twilio Programmable Messaging.
- Monitor workflow executions in n8n.
- Verify call and messaging activity through the Twilio Console.
- Understand webhook security and request-signature validation.

2. Tools & Technologies Used

n8n Cloud

Workflow automation platform used for workflow triggering, HTTP API requests, webhook reception, conditional logic, data processing, SMS API requests, and execution monitoring.

Twilio Voice API

Used to initiate the outbound phone call and connect the call to the configured TwiML/webhook instructions.

Twilio Programmable Messaging

Used to send the confirmation SMS after the recipient successfully presses the required keypad option.

TwiML

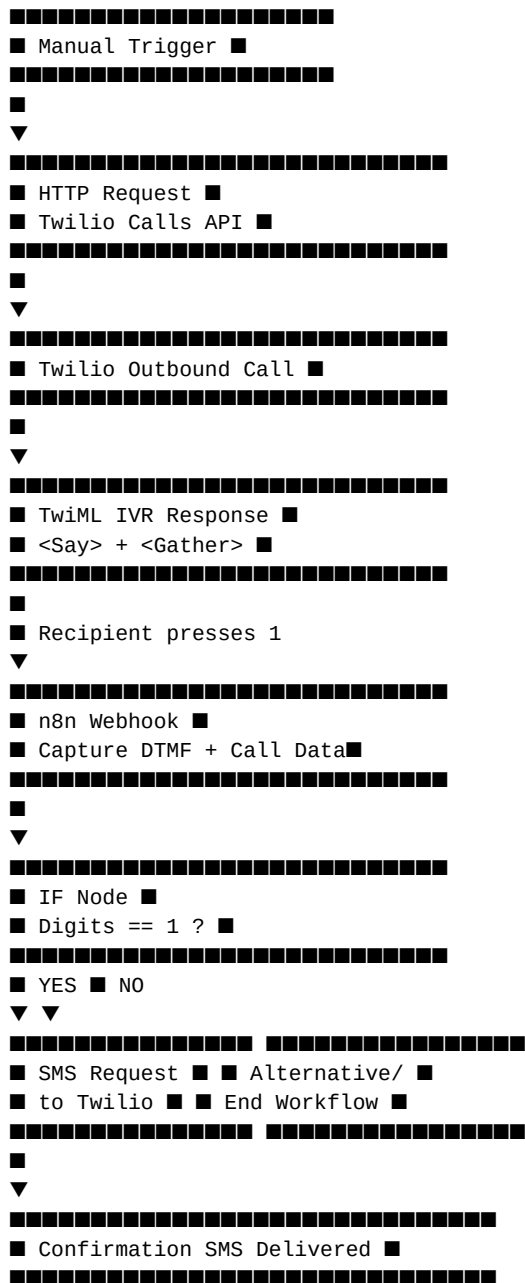
Twilio Markup Language, an XML-based instruction format used to tell Twilio what to do during a call.

Webhooks

n8n webhook endpoints provide the communication point through which Twilio sends information such as keypad input, Call SID, and caller information.

3. Workflow Architecture & Step-by-Step Flow

Overall sequence: Manual Trigger → HTTP Request → Twilio Outbound Call → TwiML IVR → DTMF Input → n8n Webhook → IF Node → SMS Confirmation



3.1 Manual Trigger

The workflow starts with an n8n Manual Trigger node, allowing the workflow to be tested manually from the n8n editor.

3.2 HTTP Request for Outbound Call

An n8n HTTP Request node communicates with the Twilio Calls API using HTTP POST, Basic Authentication, the Account SID/Auth Token, destination phone number, Twilio caller number, and the TwiML/webhook URL.

3.3 TwiML IVR Response

After the call is connected, Twilio requests instructions from the configured TwiML/webhook URL. The response greets the recipient and asks them to press 1.

3.4 Capture DTMF Input

The **<Gather>** TwiML verb collects one DTMF digit and submits it to the configured webhook. Relevant values include Digits, CallSid, From, and To.

3.5 IF Node

The IF node evaluates the received keypad input. If Digits equals 1, the workflow continues to the SMS confirmation branch.

3.6 SMS Confirmation

The workflow sends an HTTP request to the Twilio Messaging API. The SMS can contain: "Thank you! Your confirmation has been received successfully."

4. Credential Configuration

The workflow requires the following Twilio information: Account SID, Auth Token, and Twilio Phone Number. These credentials are obtained from the Twilio Developer Console.

4.1 Account SID

The Account SID uniquely identifies the Twilio account/project and is required when making authenticated API requests.

4.2 Auth Token

The Auth Token is a sensitive authentication credential. It must be kept private and should not be published on GitHub, included directly in screenshots, embedded in public documentation, or shared in demonstrations. Credentials should be stored using n8n's credential management system.

4.3 Twilio Phone Number

The Twilio phone number acts as the caller/sender identity. For voice calls it is used as the From value; for SMS it is used as the sender where supported.

5. TwiML IVR Design

The IVR is implemented using TwiML. A simplified design is shown below.

```
<Response>
<Say>
Hello. Press 1 to confirm your appointment.
</Say>

<Gather
numDigits="1"
method="POST"
action="YOUR_N8N_WEBHOOK_URL">
</Gather>
</Response>
```

5.1 <Response>

<Response> is the root element of a TwiML response and contains the instructions that Twilio should execute during the call.

5.2 <Say>

<Say> converts text into speech. It is used to provide the IVR greeting and instruction.

5.3 <Gather>

<Gather> collects input from the caller. In this project it collects one DTMF digit and sends the collected input to the n8n webhook.

5.4 DTMF Processing

DTMF stands for Dual-Tone Multi-Frequency. It is the signaling method used when a caller presses keys on the telephone keypad.

```
Caller presses 1
↓
Twilio captures DTMF
↓
Twilio sends webhook request
↓
n8n receives Digits=1
↓
IF Node evaluates the value
↓
SMS confirmation
```

6. Webhook Processing & Security

6.1 Webhook Processing

The n8n Webhook node acts as the receiving endpoint for Twilio. The workflow extracts relevant information such as Digits, CallSid, and From.

6.2 X-Twilio-Signature

Twilio can include an **X-Twilio-Signature** HTTP header with webhook requests. The signature provides a mechanism for verifying that the request originated from Twilio and was not modified during transmission.

6.3 Why Signature Validation Matters

Webhook endpoints are publicly accessible URLs in many automation architectures. Without request validation, an unauthorized party could potentially send forged requests to the webhook. This could cause unwanted workflow execution, unauthorized SMS messages, fake keypad responses, or unnecessary account-resource consumption.

6.4 Security Best Practices

- Never expose the Auth Token publicly.
- Store credentials in n8n Credentials.
- Do not commit secrets to GitHub.
- Validate Twilio webhook signatures in production.
- Use HTTPS webhook endpoints.
- Restrict access where appropriate.
- Avoid publishing personally identifiable phone numbers unnecessarily.

7. Known Limitations

7.1 Twilio Trial Account Restrictions

Twilio trial accounts have restrictions compared with fully upgraded accounts. SMS delivery to certain destinations, including +91 Indian numbers, can be affected by trial-account restrictions and account configuration. Exact behavior depends on the Twilio account, destination, sender configuration, and current Twilio policies.

7.2 Error 572006

During SMS testing, an error such as 572006 may be encountered. This indicates a Twilio messaging delivery/configuration issue rather than necessarily an error in the n8n workflow logic.

7.3 Handling the Limitation

- Verify the Twilio Console error details.
- Confirm that the destination number is supported.
- Check whether the recipient number must be verified under the trial account.
- Verify the Twilio sender configuration.
- Review Twilio Messaging logs.
- Upgrade/configure the Twilio account if required.
- Capture the Twilio error as evidence in the documentation.

For assessment purposes, the workflow execution and error details should be documented honestly rather than presenting an unsuccessful SMS delivery as successful.

8. Execution Monitoring & Verification

8.1 n8n Execution Monitoring

The n8n execution history is used to verify that each workflow stage executed correctly.

Step 1 — Trigger

Confirm that the Manual Trigger started the workflow successfully.

Step 2 — Outbound Call

Verify that the HTTP Request node successfully submitted the call request to Twilio.

Step 3 — IVR

Confirm that the call connected and the IVR greeting was played.

Step 4 — DTMF

Press 1 and verify that the webhook receives Digits = 1.

Step 5 — Conditional Routing

Confirm that the IF node evaluates the received value correctly and follows the TRUE branch.

Step 6 — SMS

Verify that the SMS request was sent to Twilio.

Step 7 — Final Execution

Review the n8n execution status and confirm completion.

Manual Trigger
↓

HTTP Request
↓
Outbound Call
↓
IVR
↓
DTMF = 1
↓
Webhook
↓
IF = TRUE
↓
SMS Request
↓
Execution Complete

9. Twilio Console Verification

The Twilio Console provides a second verification layer. Review voice call activity, call status, Call SID, messaging activity, message status, error codes if any, and request/response information where available.

10. Assessment Evidence

Twilio Console Screenshots

- Twilio project dashboard
- Account SID section
- Twilio phone number
- Voice call activity
- Messaging activity
- Relevant execution or error details

n8n Screenshots

- Complete workflow
- Manual Trigger
- HTTP Request node
- Webhook node
- IF node
- SMS HTTP Request node
- Successful execution
- Webhook execution data
- Captured Digits value

IVR Evidence

- TwiML configuration
- IVR prompt
- configuration

- Keypad interaction

SMS Evidence

- SMS request execution
- Twilio message record
- Delivery status where available

11. End-to-End Workflow Summary

- The user manually starts the n8n workflow.
- n8n sends an HTTP request to the Twilio Calls API.
- Twilio initiates the outbound phone call.
- Twilio requests the configured TwiML instructions.
- The IVR uses **<Say>** to greet the recipient.
- **<Gather>** asks the recipient to press 1.
- The recipient presses 1.
- Twilio captures the DTMF input.
- Twilio sends the captured information to the n8n webhook.
- n8n extracts the keypad digit and call information.
- The IF node checks whether the digit is 1.
- If the condition is true, n8n sends an SMS request to Twilio.
- Twilio processes the SMS.
- The n8n execution history records the workflow execution.
- The Twilio Console is used to verify call and messaging activity.

12. Conclusion

The Twilio IVR Call & SMS Confirmation Automation demonstrates an end-to-end integration between n8n workflow automation and Twilio communication APIs. The project covers REST API integration, API authentication, voice automation, TwiML, Interactive Voice Response, DTMF input handling, webhooks, conditional workflow routing, SMS automation, execution monitoring, and webhook security.

The workflow provides a practical example of how voice interactions can be connected to automated backend processes and follow-up messaging. It also demonstrates the importance of handling external service limitations, monitoring API responses, protecting authentication credentials, and validating webhook requests in production environments.

13. Final Assessment Checklist

Assessment Item	Status
Twilio Console Accessed	■
Account SID Retrieved	■
Auth Token Configured	■
Twilio Phone Number Configured	■

Assessment Item	Status
n8n Workflow Created	■
Outbound Call Workflow Built	■
Twiml IVR Created	■
Configured	■
Configured	■
DTMF Input Captured	■
n8n Webhook Configured	■
IF Node Configured	■
SMS Confirmation Workflow Built	■
Workflow Execution Verified	■
Twilio Console Logs Reviewed	■
Webhook Security Explained	■
Known Limitations Documented	■
Screenshots Added	■
Video Demonstration Recorded	■
GitHub Repository Prepared	■

14. Submission Deliverables

1. Documentation

- Workflow architecture
- Configuration details
- Twiml design
- Webhook processing
- Security explanation
- Limitations
- Execution evidence
- Screenshots

2. Video Demonstration

- Twilio Console
- Credentials/configuration
- n8n workflow
- Outbound call
- IVR prompt
- Keypad interaction
- Webhook execution
- SMS workflow

- Execution logs

3. GitHub Repository

```
twilio-ivr-n8n/  
■  
■■■ README.md  
■■■ workflow/  
■ ■■■ n8n-workflow.json  
■  
■■■ twiml/  
■ ■■■ ivr.xml  
■  
■■■ documentation/  
■ ■■■ project-report.pdf  
■  
■■■ screenshots/  
■ ■■■ n8n-workflow.png  
■ ■■■ webhook-execution.png  
■ ■■■ twilio-console.png  
■ ■■■ call-log.png  
■ ■■■ sms-log.png  
■  
■■■ evidence/  
■■■ execution-summary.md
```

Security reminder: Never upload the Twilio Auth Token or any other secret credential to a public repository.